

Stanley Lew

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Education

University of California - Riverside

Riverside, CA

BS in Computer Science (3.85 GPA)

September 2023 - June 2027

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, HTML, CSS, SQL

Frameworks & Technologies: React, React Native, Next.js, Expo, Vite, Supabase, Firebase, TailwindCSS, NativeWind, TensorFlow, Resend, BeautifulSoup, ZeroMQ

Developer Tools: Git, GitHub, GitHub Actions, Google Cloud Platform, Ollama, CMake, Jest, GoogleTest

Project Management: Jira, Agile, Scrum, Kanban, GitHub Projects

Work & Leadership Experience

Collaborative Intelligence Systems Lab

Riverside, CA

Undergraduate Researcher

September 2025 - Present

- Built automated stereo-camera trajectory generation in Habitat-Sim (headless) for simultaneous localization and mapping (**SLAM**) data collection in Coop3rSLAM
- Implemented an asynchronous **ZeroMQ-based** communication layer with typed packet enums and serialization for multi-agent message passing

Association for Computing Machinery @ UCR (ACM@UCR)

Riverside, CA

Projects Director, Web Development Director, & Project Lead

February 2024 - Present

- **Managed 50+ Next.js and 5+ Figma projects** and hosted weekly check-in meetings with project leads to oversee development of websites for UCR's CSE department, student organizations, & external companies
- **Increased ACM programs membership by over 150%**, driving engagement and participation through effective outreach and project-based learning opportunities for 100+ members quarterly
- Led 5+ teams of 10+ students as **Scrum Master**, facilitating weekly sprint planning, code reviews, and **AGILE** development

Projects

SVG Lucide Icon Finder | *TypeScript, HTML5, TailwindCSS, Tensorflow, Vite*

🐙 github.com/stanleylew5/svg-lif

- Developed a Chrome extension that converts user-provided SVGs into normalized tensors and uses **TensorFlow.js** with cosine similarity to identify the closest-matching Lucide React icon
- Implemented in-memory **caching** of preprocessed icon tensors, reducing repeated SVG-to-tensor conversions and improving match performance by about **50%** on subsequent searches.

SWE AI Agent | *Python, Ollama*

🐙 github.com/acm-ucr/swe-agent

- Implemented core SWE-agent features to autonomously read, edit, and write code based on task prompts using models on **Ollama**, simulating real-world software engineering workflows
- Engineered GitHub workflow automation for the AI agent to autonomously create pull requests, resolve merge conflicts, and merge changes into the target repository.
- Engineered a task hierarchy and inter-device communication system using **ZeroMQ** and **JSON messaging** to coordinate actions between distributed components

Tethr | *TypeScript, NativeWind, Supabase, Expo, Jest*

🐙 github.com/stanleylew5/tethr

- Implemented task and group controller logic following an **MVC architecture**, connecting user-group-task workflows and ensuring clean separation between views, controllers, and data model
- Built a **CI/CD** pipeline using **GitHub Actions** to automatically enforce formatting, linting, type checking, builds, and test execution on pull requests
- Designed and debugged Supabase Row Level Security (**RLS**) **policies** across interrelated tables (users, groups, tasks), ensuring secure access control and correct permission handling

R'Eats | *Python, BeautifulSoup, Next.js, TypeScript, TailwindCSS, Github Actions*

🐙 github.com/stanleylew5/r-eats

- Developed a **Python** web scraping script with **BeautifulSoup** to extract and structure dining menu data into JSON for easier consumption
- Improved student access to dining information, attracting **500+** unique visitors on the first day and **200+** unique daily users since launch